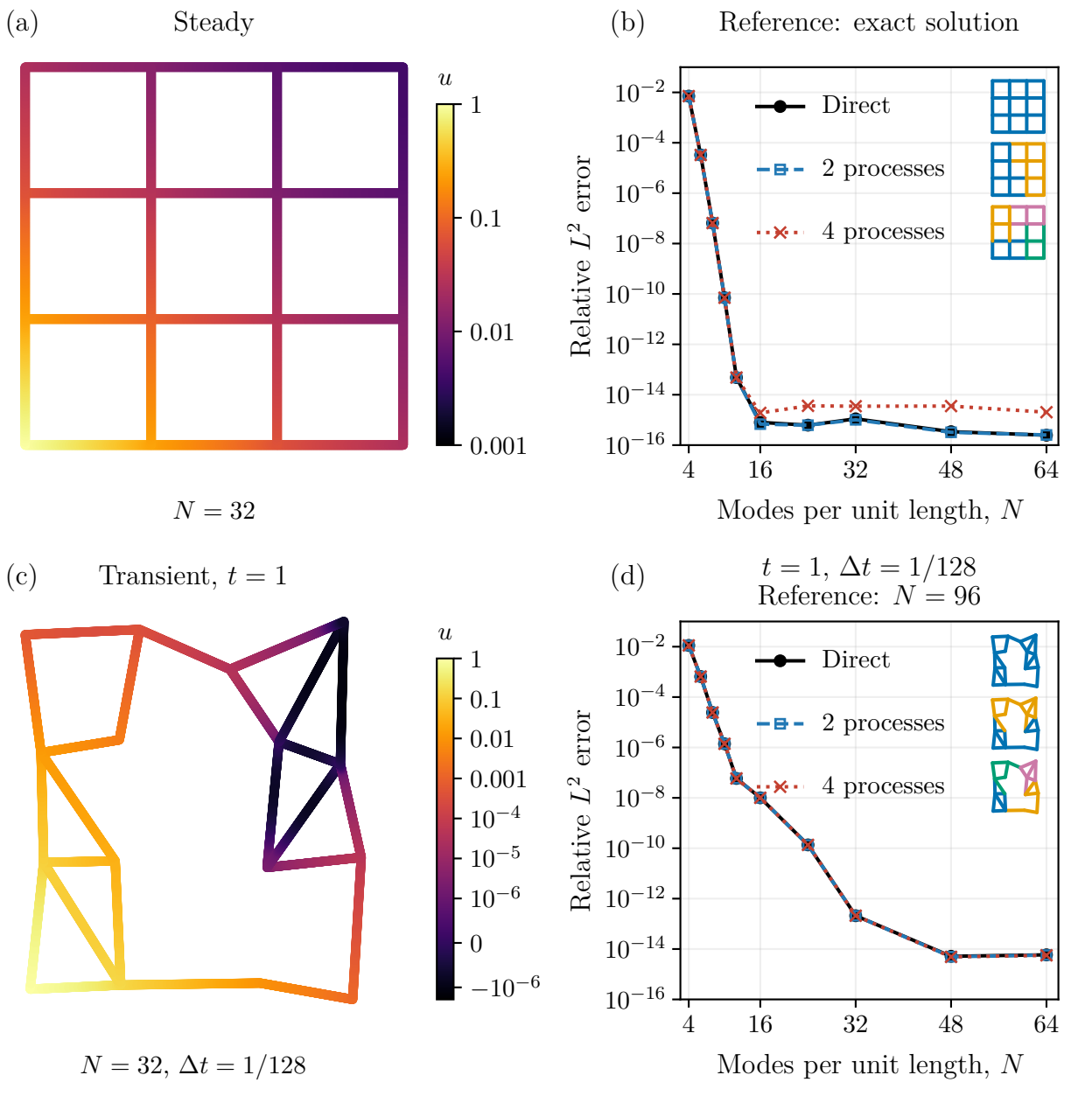

(a)
Steady
u
1
0.1
0.01
0.001
N = 32
(b)
Reference: exact solution
Direct
2 processes
4 processes
Relative L^2 error
Modes per unit length, N
(c)
Transient, t = 1
u
1
0.1
0.01
0.001
10^-4
10^-5
10^-6
0
−10^-6
N = 32, Δt = 1/128
(d)
t = 1, Δt = 1/128
Reference: N = 96
Direct
2 processes
4 processes
Relative L^2 error
Modes per unit length, N